

Lab #02: Fun with ASCII

Purpose

The general purpose of this laboratory is to familiarize you further with rotate, shift, and logical operations as well as conditional execution.

You are NOT allowed to use branches in your solution.

The ONLY instructions that you are allowed to use in your solution are:

- Rotates
- Shifts
- Moves
- Compares
- Logical (AND, ORR, etc)
- Mathematical (ADD, SUB, MUL, xDIV)

In addition, your entire program including comments, empty lines, and directives must be 150 lines or less. Also, you are NOT allowed to remove any lines from the starter file before line 26. You can only add lines.

Instructions

Use the starter project provided to you by the instructor. Submit the project folder to Canvas in ONE zip file.

Overview

The objective of this program is to count specific ASCII characters in a given binary pattern. You will be looking for the ASCII character 0-9, A-O, and the NULL character. Refer to an ASCII table for the 8-bit binary pattern for all characters. Only those characters will appear in the pattern. You do not have to worry about any other ASCII codes.

Task 1

Your code will analyze the given pattern and count how many of the ASCII characters refer to the characters 0-9. You will need to calculate two things:

1. SUM: [R3] the sum of all numbers appearing in the binary string (you are summing the number that the code represents, not the code itself)
2. Count: [R4] how many numbers (0-9) appear in the binary string

Task 2

Your code will analyze the given pattern and count how many particular ASCII characters are present in the binary string. You will need to calculate two things:

1. Letter count: [R5] the number of letter from A-O (capital only) that appear in the binary string
2. NULL count: [R6] how many times the ASCII code for NULL appears in the binary string

Testing & Debugging

You are provided some testing binary strings in the starter file. DO NOT remove those. These will be used to test and help you debug your code.

Submission (through Canvas Assignments)

Submit the complete and working project to Canvas as one zip file.

- The project file will be named Lab2.uvprojx
- The project folder will be named ECE_3436_Lab2_Team##.
- Therefore, the zip file will be named ECE_3436_Lab2_Team##.zip.

Keep your program (the entire main.s file) to 150 lines or less!

ASCII Table

ASCII (1977/1986)																
	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0x	NUL	SOH	STX	ETX	EOT	ENQ	ACK	BEL	BS	HT	LF	VT	FF	CR	SO	SI
1x	DLE	DC1	DC2	DC3	DC4	NAK	SYN	ETB	CAN	EM	SUB	ESC	FS	GS	RS	US
2x	SP	!	"	#	\$	%	&	'	(	)	*	+	,	-	.	/
3x	0	1	2	3	4	5	6	7	8	9	:	;	<	=	>	?
4x	@	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
5x	P	Q	R	S	T	U	V	W	X	Y	Z	[	\	]	^	_
6x	`	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o
7x	p	q	r	s	t	u	v	w	x	y	z	{		}	~	DEL
Changed or added in 1963 version Changed in both 1963 version and 1965 draft																